



CSL 252- Design and Analysis of Algorithms
Indian Institute of Technology Bhilai
Tutorial Sheet 3

1. What are the minimum and maximum numbers of elements in a heap of height h ?
2. Show that an n -element heap has height $\lfloor \log n \rfloor$.
3. Show that in any subtree of a max-heap, the root of the subtree contains the largest value occurring anywhere in that subtree.
4. Where in a max-heap might the smallest element reside, assuming that all elements are distinct?
5. Is an array that is in sorted order a min-heap?
6. Is the array with values $\{23, 17, 14, 6, 13, 10, 1, 5, 7, 12\}$ a max-heap?
7. Show that, with the array representation for storing an n -element heap, the leaves are the nodes indexed by $\lfloor \frac{n}{2} \rfloor, \lfloor \frac{n}{2} \rfloor + 1, \lfloor \frac{n}{2} \rfloor + 2, \dots, n$.
8. Starting with the procedure MAX-HEAPIFY, write pseudocode for the procedure MIN-HEAPIFY(A, i), which performs the corresponding manipulation on a min-heap. How does the running time of MIN-HEAPIFY compare to that of MAX-HEAPIFY?
9. What is the effect of calling MAX-HEAPIFY(A, i) when the element $A(i)$ is larger than its children?
10. Show that the worst-case running time of MAX-HEAPIFY on a heap of size n is $\Omega(\log n)$. (Hint: For a heap with n nodes, give node values that cause MAX-HEAPIFY to be called recursively at every node on a simple path from the root down to a leaf.)
11. Show that there are at most $\lceil \frac{n}{2^{h+1}} \rceil$ nodes of height h in any n -element heap.
12. Let A and B be two arrays of integers of size n and $2n$, respectively. Design an $O(n \log n)$ algorithm that returns "TRUE" if $A \subseteq B$, otherwise, returns "FALSE".
13. Describe an $O(n \log n)$ time algorithm that, given an array A of n integers and another integer x , determines whether or not there exist two integers in A whose sum is exactly x .
14. Why do we want the loop index i in line 2 of BUILD-MAX-HEAP to decrease from $\lfloor \frac{A.length}{2} \rfloor$ to 1 rather than increase from 1 to $\lfloor \frac{A.length}{2} \rfloor$?
15. Illustrate the operation of PARTITION on the array $\{13, 19, 9, 5, 12, 8, 7, 4, 21, 2, 6, 11\}$.
16. What value of q does PARTITION return when all elements in the array $A[p \dots r]$ have the same value?
17. How would you modify QUICKSORT to sort into non-increasing order?
18. What is the running time of QUICKSORT when all elements of array A have the same value? Show that the running time of QUICKSORT is $O(n^2)$ when the array A contains distinct elements and is sorted in decreasing order.



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19. Suppose that the splits at every level of quicksort are in the proportion $1 - \alpha$ to α , where $0 < \alpha < \frac{1}{2}$ is a constant. Show that the minimum depth of a leaf in the recursion tree is approximately $-\frac{\log n}{\log \alpha}$ and the maximum depth is approximately $-\frac{\log n}{\log(1-\alpha)}$ (Don't worry about integer round-off.)
20. Give an $O(n \log k)$ -time algorithm to merge k sorted lists into one sorted list, where n is the total number of elements in all the input lists.
21. Let A be an array of n integers. Describe an algorithm that finds the maximum and minimum of the array A with at most $3\lceil \frac{n}{2} \rceil$ comparisons.
22. A sorting algorithm is called stable, if for any input array A , the relative position of two equal keys will not change before and after the sorting. For example, if $a[2]$ and $a[5]$ have the same value 9 in the input array, then the key 9 of $a[2]$ will appear before the key 9 of $a[5]$ in the sorted array. Example of some stable sorting algorithms are Quick sort, Merge Sort.
- (a) Is Heap sort stable? Explain your answer.
 - (b) Show that any sorting algorithm can be made stable without increasing its asymptotic time complexity.